Release-v4 explanatory supplement (E04)

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1 Compensated response and likelihood-product explanations

These finite-horizon explanations expand the retained manuscript identities; they do not replace the scientific procedures or establish a stationary nonlinear exact-memory theorem.

1.1 Compensator cancellation

With physical productivity $\mathbb{E}\kappa = n$, scalar initialization $X_0\Phi$, zero initial excitation and no additional prehistory, set $f = X_0\Phi + \rho(1 * \Phi)$ and write X for post-event stress in ordinary time convolutions. Then $X = f - \Phi * \mathrm{d}N_g$ and $\lambda = \Psi(X_-) + h * \mathrm{d}N_\kappa$. The deterministic resolvent identities are

$$Q_\gamma + nR_\gamma * h = R_\gamma, \qquad Q_\gamma + \gamma R_\gamma * \Phi = \Phi.$$

Substituting $\mathrm{d}M_g = \mathrm{d}N_g - m_1\lambda\,\mathrm{d}t$ and $\mathrm{d}M_\kappa = \mathrm{d}N_\kappa - n\lambda\,\mathrm{d}t$ into the right-hand side S of the manuscript's compensated identity gives

$$\begin{aligned} S &= X_0Q_\gamma + \rho(1 * Q_\gamma) + \gamma R_\gamma * X - m_1R_\gamma * \Psi - Q_\gamma * \mathrm{d}N_g \\ &\quad + m_1Q_\gamma * \lambda - m_1R_\gamma * h * \mathrm{d}N_\kappa + m_1nR_\gamma * h * \lambda \\ &= X_0Q_\gamma + \rho(1 * Q_\gamma) - Q_\gamma * \mathrm{d}N_g + \gamma R_\gamma * X. \end{aligned}$$

Here the first kernel relation combines the compensators, after which $\lambda - \Psi = h * \mathrm{d}N_\kappa$ cancels the remaining terms. The second relation makes the first three final terms $X - \gamma R_\gamma * X$, so $S = X$. This reasoning does not require independence of the two mark martingales. Predictable intensities still use the left limit; changes on the countable jump set do not affect ordinary time convolutions. The algebra is causal on finite horizons, without a transform assumption for the nonlinear sample path.

1.2 An event-free deterministic check

For $n = 0$, $m_1 = \ell = \gamma = \tau = X_0 = \rho = 1$ and $\alpha = 1/2$, take $\Psi(X) = X$ on the positive range of the path. Then

$$\Phi(t) = e^t \operatorname{erfc}(\sqrt{t}), \qquad X(t) = \Phi(t) + \int_0^t \Phi(u)\,\mathrm{d}u, \qquad \widehat{Q}(s) = \widehat{R}(s) = \frac{1}{s + \sqrt{s} + 1}.$$

Although no raw event occurs, $\mathrm{d}M_g = -X\,\mathrm{d}t$ and $-R * \mathrm{d}M_g = R * X$. The nonlinear residual is zero, giving

$$X = Q + 1 * Q + R * X.$$

At $t = 1$, deterministic evaluation gives $X(1) = 0.983546319407127$, $Q(1) + (1 * Q)(1) = 0.623004105263315$, and $(R * X)(1) = 0.360542214143811$, up to displayed rounding. The compensator accounts for the apparent discrepancy. Replacing only the initial/loading kernels Q by Φ instead adds $\gamma R * f$. The linear link is used only on this positive stress range, and a positive extension outside it can be supplied; no globally negative intensity is needed. These are fixed-parameter evaluations, not simulated catalogues.

1.3 The likelihood-product envelope

At fixed generating nuisances, let

$$\mathcal{B} = \int_0^T B(t)\,\mathrm{d}t, \quad J_\eta = \int_0^T B(t)(1 - e^{-\eta G_s(t)})\,\mathrm{d}t, \quad q_i = \frac{B_i e^{-\eta G_i} + c_i}{B_i + c_i}, \quad c_i = nH_i.$$

At actual reference events $B_i + c_i > 0$ almost surely. On $\eta \in [\eta_-, \eta_+] \subset (0, \infty)$, the bound $\sup_{z \geq 0} z e^{-\eta_- z} = 1/(e\eta_-)$ gives $|q_i'| \leq 1/(e\eta_-)$ and $|J_\eta'| \leq \mathcal{B}/(e\eta_-)$. As $0 \leq q_i \leq 1$ and $0 \leq J_\eta \leq \mathcal{B}$,

$$\begin{aligned} |\partial_\eta L_\eta| &\leq e^{J_\eta} \left(|J_\eta'| \prod_i q_i + \sum_i |q_i'| \prod_{j \neq i} q_j \right) \\ &\leq \frac{e^{\mathcal{B}}(\mathcal{B} + N(T))}{e\eta_-}. \end{aligned}$$

No event factor is divided out. Since $\mathbb{E}_0 N(T) \leq \mathcal{B}/(1 - n)$, this is an integrable envelope for differentiating the expectation of $L_\eta I$ with the fixed bounded rejection indicator I. It is a derivative bound for L_η, not $\log L_\eta$. It requires neither integrability of $\sum_i G_i$ nor an added stress-mark moment assumption. The existing null-set convention, fixed-rule conditions, positive-η restriction and non-Wilks limitations remain in force.

1.4 Why the small-set clock is shared

For one common Poisson measure with intensity $\mathrm{d}s\,\mathrm{d}u\,F(\mathrm{d}m)$, the no-proposal event in $[0, t] \times [0, \Lambda_R] \times \mathsf{M}$ has probability $e^{-\Lambda_R t}$ because $F(\mathsf{M}) = 1$. Both event-free flows from the specified sublevel set have rates at most Λ_R. The first accepted jump would therefore contradict this event. This justifies the retained common-clock lower bound without asserting a rate bound after arbitrary jumps.

1.5 Paired MCSE convention

For paired binary decisions let $D_i \in \{-1, 0, 1\}$ be first minus second, with b positive and c negative discordances among N_{pair} pairs. Since $\sum D_i = b - c$ and $\sum D_i^2 = b + c$,

$$\overline{D} = \frac{b - c}{N_{\mathrm{pair}}}, \qquad \widehat{\mathrm{MCSE}} = \sqrt{\frac{b + c - N_{\mathrm{pair}}\overline{D}^2}{N_{\mathrm{pair}}(N_{\mathrm{pair}} - 1)}}.$$

This is sample standard deviation divided by $\sqrt{N_{\mathrm{pair}}}$. At 150 pairs, 29/21 yields 4.709613821 percentage points and 61/6 yields 4.577609808. The empirical-variance divisor N_{pair} instead of $N_{\mathrm{pair}} - 1$ yields 4.693888856 and 4.562325593 from the same outcomes. These are conditional descriptive summaries under fixed-bank and adaptive reuse qualifications, without a new test or confidence interval.